

Executive Quality Summary

2

Observations

1

Datasets

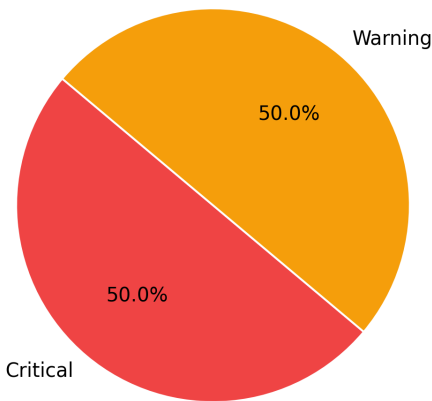
1

Critical Risks

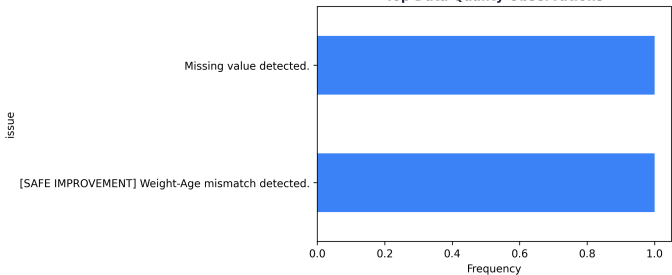
INTEGRITY SCORE

93%

Issue Severity Distribution



Top Data Quality Observations



Clinical Risk Narrative

The data audit of 1 datasets reveals a Clinical Integrity Score of 93%. This score is calculated based on the density of critical clinical outliers (values outside physiological norms) and data consistency issues. High-risk anomalies were detected that could potentially compromise clinical research modeling. Analysis suggest that protocol deviations or legacy data entry methods are the primary drivers of these observations.

Dataset Name	Total Issues	Critical Issues
dmsap_test.csv	2	1

Clinical Intelligence Deep-Dive

This section highlights the most significant clinical findings analyzed by our AI system. Unlike raw validation, this deep-dive provides clinical context, impact, and actionable remediation steps.

FINDING #1: Body_Temp | dmsap_test.csv

****DETAILED OBSERVATION****

The missing Body_Temp value for Patient_ID P007 is suspicious because temperature is a critical measurement in clinical contexts, affecting patient outcomes and treatment decisions. Without accurate temperature data, downstream analyses could lead to misdiagnosis or incorrect treatment.

****RECOMMENDED ACTION****

Set the missing value as 'nan' since it's already present. This ensures consistency without additional interpolation unless more data is available.

Detailed Audit Logs (Sample)

DATASET	COLUMN	OBSERVATION	SEV	STATUS
dmsap_test.csv	Body_Temp	Missing value detected.	Cri	AUDITED
dmsap_test.csv	Wt	[SAFE IMPROVEMENT] Weight-Age mismatch detected. (Weigh	War	AUDITED

Strategic Roadmap & Methodology

Audit Methodology:

The audit utilized a dual-engine validation process. Phase 1 involved deterministic clinical rules based on DMSAP (Data Management & Statistical Analysis Plan) standards. Phase 2 leveraged a local DeepSeek-R1 LLM to analyze row context and provide actionable clinical intelligence. All PII was handled according to masking protocols enabled during the session.

Short-term (0-30 days)

Immediate cleanup of 'Critical' flags identified in Section 4. Standardize field headers across all source datasets.

Mid-term (30-90 days)

Develop automated ETL pipelines with pre-processing gates for temperature and age outliers.

Long-term (Strategic)

Adopt CDISC SDTM standards for all clinical research datasets to ensure global interoperability and compliance.